Dense Retrieval Basics

1 Introduction

Dense retrieval is a method that finds passages by comparing embedding vectors of the query and the passages. A bi-encoder is a model that encodes the query and each passage separately into vectors. Cosine similarity is a measure of the angle between two embedding vectors.

2 Strengths and costs

Dense retrieval captures meaning that shares no words with the query. Dense retrieval needs an index of vectors built ahead of time. Approximate nearest neighbour search makes dense retrieval fast on large collections. Dense retrieval can miss rare names and acronyms that a keyword method finds easily.